Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

Roll No: 2601188
Name: Sakshi Rajendra Suryawanshi

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#singly linked list : 10 oct 2026


class Node:
    def __init__(self,value):
        self.data = value
        self.next = None

class SLL:
    def __init__(self):
        self.head = None

    #append a Node
    def append(self,new_node):
        if self.head == None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next:
                temp = temp.next
            temp.next = new_node

     #display all nodes
    def display(self):
        temp = self.head
        while temp:
```

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        print(temp.data)
        temp = temp.next

# Insert a node at a 1-based position
    def insert(self,new_node,pos):
        if(pos == 1):
            new_node.next = self.head
            self.head = new_node
        else:
            p=1
            temp = self.head
            while p != pos-1:
                temp = temp.next
                p +=1
            new_node.next = temp.next
            temp.next = new_node

     #delete  the node
    def delete(self,value):
        temp = self.head
        if(temp.data == value): #first node to delete
            self.head = self.head.next
        else:
            while(temp.data != value and temp):
                prev = temp
                temp = temp.next
                if temp ==None:
                    print("Value isn't present")
                    return
                prev.next = temp.next
                temp = None
```

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#find middle node
def findMiddle(self):
    p1 = self.head
    p2 = self.head

    while p2 and p2.next:
        p1 = p1.next
        p2 = p2.next.next

    if p1:
        print(p1.data)
    else:
        print("list is empty")

#Reverse the List
def reverse(self):
        prev = None
        current = self.head
        while current :
            next_node = current.next
            current.next = prev
            prev = current
            current = next_node
        self.head = prev

#Calculate the sum of every two consecutive node values
def SumOfNode(self):
    current = self.head
    while current and current.next:
        sum = current.data + current.next.data
        print(sum)
```

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            current = current.next

#Main Program
ll = SLL()

n1 = Node(10)
n2 = Node(20)

print("Appended list :")
ll.append(n1)
ll.append(n2)
ll.display()

print()
print("Inserted List :")
ll.insert(Node(30),1)
ll.insert(Node(50),3)
ll.display()

print("deleted Node :")
ll.delete(30)
ll.display()



print("Reverse List :")
ll.reverse()
ll.display()

print()
print("Sum of 2 consecutive nodes : ")
ll.SumOfNode()
```

```
print("Middle value:")
ll.findMiddle()
```

Sample Output

```
Appended list:

  10
  20

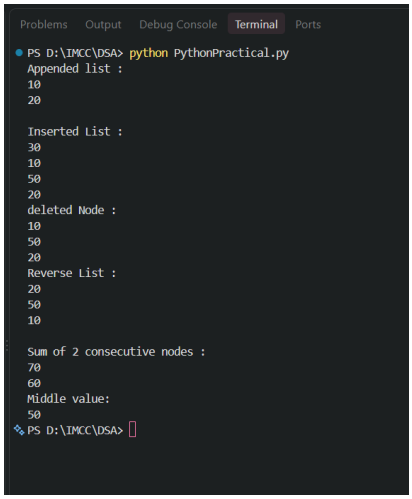
  Inserted list:
  30
  10
  50
  20

  Deleted node (30):
  10
  50
  20

  Reversed list:
  20
  50
  10

  Sum of 2 consecutive nodes:
  70
  60

  Middle value:
  50
```


Problems
Output
Debug Console
Terminal
Ports
PS D:\IMCC\DSA> python PythonPractical.py
Appended list :
10
20
Inserted List :
30
10
50
20
deleted Node :
10
50
20
Reverse List :
20
50
10
Sum of 2 consecutive nodes :
70
60
Middle value:
50
PS D:\IMCC\DSA>